

Day 22 Student Handout

Geometry Summer Credit | Scale Effects, Density, and Design Constraints

Name:	Date:	Period:
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Learning Targets

I can describe how scale factor affects length, area, and volume. I can solve density and design-constraint problems using geometry formulas. I can explain why units matter in modeling.

Lesson 22A: Scale Effects

Warm-Up

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| 1. | A rectangle has side lengths 5 and 8. If every side length is doubled, what are the new side lengths? |
| 2. | Find the area of the original rectangle and the doubled rectangle from item 1. |
| 3. | A cube has side length 3. Find its volume. Then find the volume if the side length is doubled. |

Scale Factor Pattern

If lengths are multiplied by scale factor k , perimeters and lengths multiply by k , areas multiply by k^2 , and volumes multiply by k^3 .

Problem	Work / notes
Two similar rectangles have scale factor 3 from smaller to larger. The smaller perimeter is 18. Find the larger perimeter.	Length/perimeter scale: New perimeter:
Two similar figures have scale factor 4. The smaller area is 20 square units. Find the larger area.	Area scale: New area:
Two similar solids have scale factor 2. The smaller volume is 45 cubic units. Find the larger volume.	Volume scale: New volume:

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Lesson 22B | Density and Design Constraints

Modeling With Geometry

Density and Constraints

Density = mass / volume. A design constraint is a limit or requirement, such as maximum cost, minimum capacity, fixed surface area, or required weight.

Problem	Work / notes
A block has mass 240 grams and volume 80 cubic centimeters. Find its density.	Formula: Substitute: Density and units:
A rectangular tank is 10 cm by 8 cm by 6 cm. Water has density 1 g/cm ³ . Find the mass of water when full.	Volume: Density relationship: Mass:
A box must hold at least 500 cubic inches. One design is 10 in by 8 in by 6 in. Does it meet the constraint?	Volume: Compare to constraint: Decision:

Choosing the Right Measurement

4.	For each situation, choose length, area, volume, or density: wrapping a gift, filling a pool, comparing heaviness for the same size, edging a garden.
5.	Explain why a model answer should include both a number and a unit.